Prioritizing Cancer Drug Targets with Monte Carlo Simulations of Network Diffusion

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ABSTRACT

Twist1, a transcription factor, plays a pivotal role in breast cancer metastasis. Unfortunately, like many potential cancer targets, it is not a druggable target. In a previously published study,[1] we used network perturbation theory to help select candidate drug targets downstream of Twist1. Here, we extend that work by using Monte Carlo simulations of network diffusion to prioritize potential downstream targets. The code is flexible, easy to understand, and easy to extend. We pay particular attention to examining how different weightings of individual nodes can affect model performance.

AIMS

- Objective: To model and analyze the Twist 1 gene network, which is implicated in breast cancer metastasis.
- Method: Utilize Monte Carlo simulations to map the gene graph network of Twist 1, focusing on different paths influenced by the gene.
- Development Focus: Implement "weighted walkers" in simulations to reflect real-world probabilities of protein interactions, acknowledging that not all interactions are consistent or frequent.

METHODS

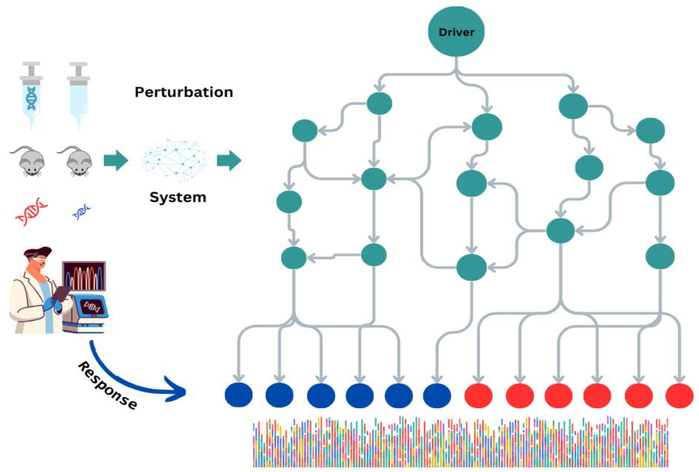

Figure 1: Schematic of System Response to Gene Editing. This diagram illustrates the process of gene editing as a perturbation in a biological system, tracing the subsequent effects through a network that culminates in diverse responses.

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RESULTS

Node Graph

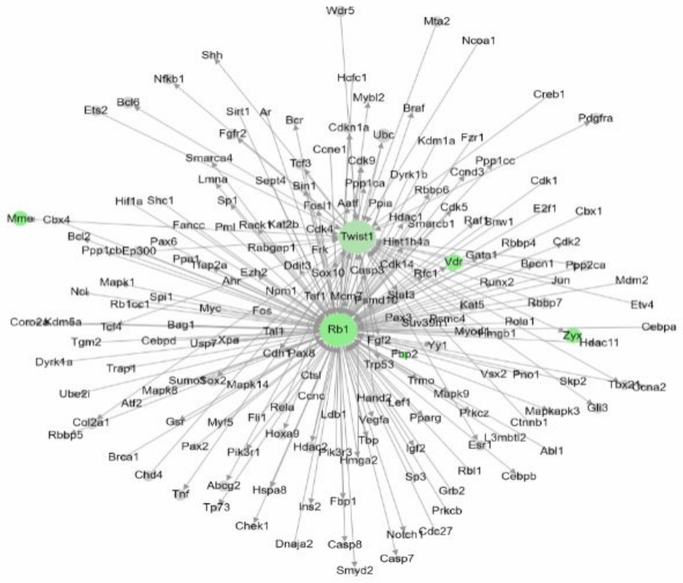


Figure 2: Comprehensive Network Map: This figure displays a detailed network map showing a balanced 1:1 ratio of protein-protein interactions to transcription factor relationships, visually capturing the intricate connections and functional dynamics within the biological system

Effect of Weighting PPIs vs TFs

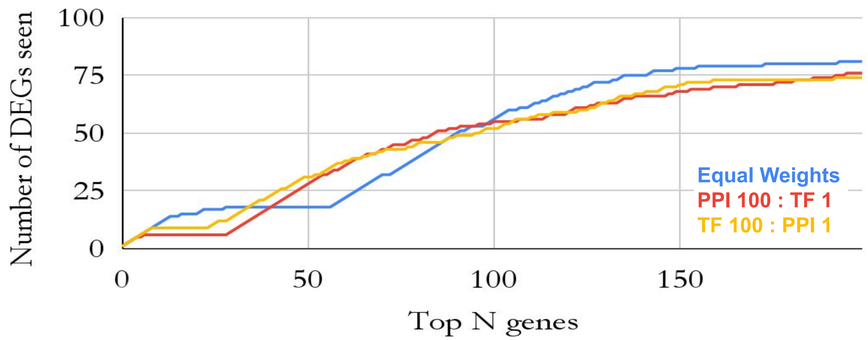
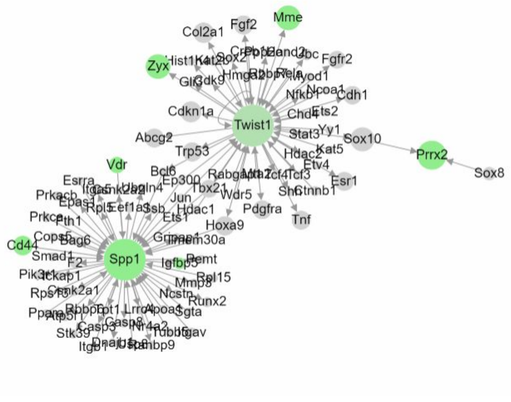


Figure 4: The effects of weighting PPIs versus TFs on the identification of differentially expressed genes (DEGs). The blue line represents the normal weighting condition, the red line shows the impact of heavily weighting PPIs (PPI 100: TF 1), and the orange line illustrates the results with a strong emphasis on TFs (TF 100: PPI 1). Each curve indicates the number of DEGs detected as more genes are considered, demonstrating the influence of weighting schemes on gene prioritization in biological datasets.

Figure 5: This table showcases the top 10 prominent nodes within a biological network, detailing their interaction counts and neighboring connections, and categorizing them by node and gene types. Nodes like 'Twist1', 'Rb1', and 'Prrx2' are featured, highlighting their functions as drivers or responders. The table also provides a breakdown of their associated TF and PPI counts, offering insights into each gene's role and impact on network connectivity and functional hierarchy.

	Node	Count	Neighbors	NodeType	GeneType
1	Twist1	12595	43	Driver	{'tf': 25, 'ppi': 24}
2	Rb1	1837	148	Response	{'tf': 16, 'ppi': 135}
…					
10	Prrx2	612	2	Response	{'tf': 0, 'ppi': 2}

Weighted Walker TF: PPI (100:1)



Weighted Walker PPI:TF (100:1)

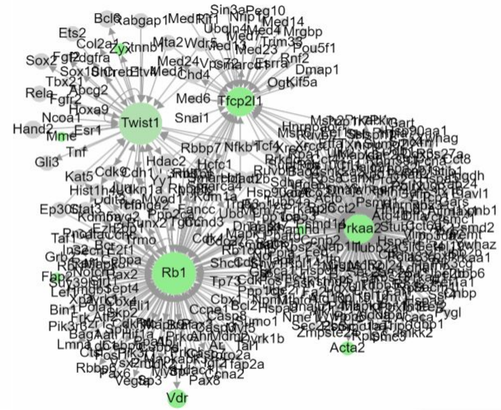


Figure 3a: Weighted Walker TF-PPI (100:1) showing a network with a high emphasis on transcription factor influence, with key nodes like 'Twist1', 'Prx2', and 'Sox8' highlighted, indicating a focused analysis on transcriptional regulation.
Figure 3b: Weighted Walker PPI-TF (100:1) displaying a network where protein-protein interactions are heavily weighted, resulting in a denser and more interconnected network visualization. Central nodes like 'Rb1' and 'Vdr' are prominent, reflecting significant roles in protein interactions.

CONCLUSIONS

- Findings: Adjusting the weight between protein-protein interactions (PPI) and transcription factors (TF) significantly alters the network's structure and dynamics.
- Insights: Key nodes and pathways affected by these weight adjustments could serve as potential therapeutic targets.
- Future Direction: Integrate more experimental data to enhance the predictive power of the network models and further understand the biological roles of Twist 1 in cancer mechanisms.

REFERENCES

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